

O

LV.0 → LV.1 · AI ENGINEER

10 O-words that move you from **Level 0** → 1.

Overfitting, Optimizers, Orchestration & more. **Plain-English** definitions, one swipe each, plus the O-named tools and minds to follow.

↓ UP NEXT · TERMS

The 0 stack.

Overfitting

Optimizer

Orchestration

Outliers

ODE

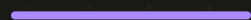
Orthogonality

One-shot Prompting

Open-weight Models

OCR

Observability



OVERFITTING • GENERALIZATION

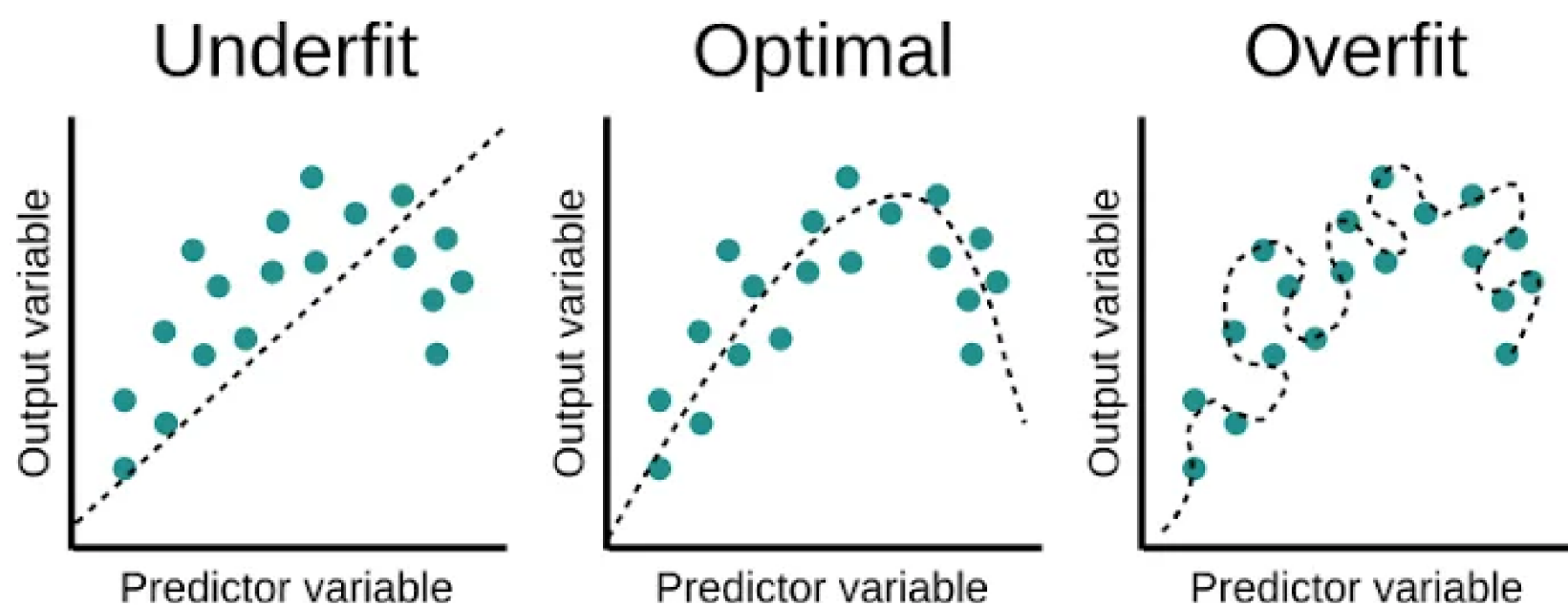
Overfitting

memorizes noise • fails on new data

When a model learns training data too precisely, including its noise, and **fails to generalize to new data**. High training accuracy, poor real-world results.

USED IN

- Model validation
- Regularization
- Train/test splits



UNDERFIT • GOOD FIT • OVERFIT

OUTLIERS • DATA

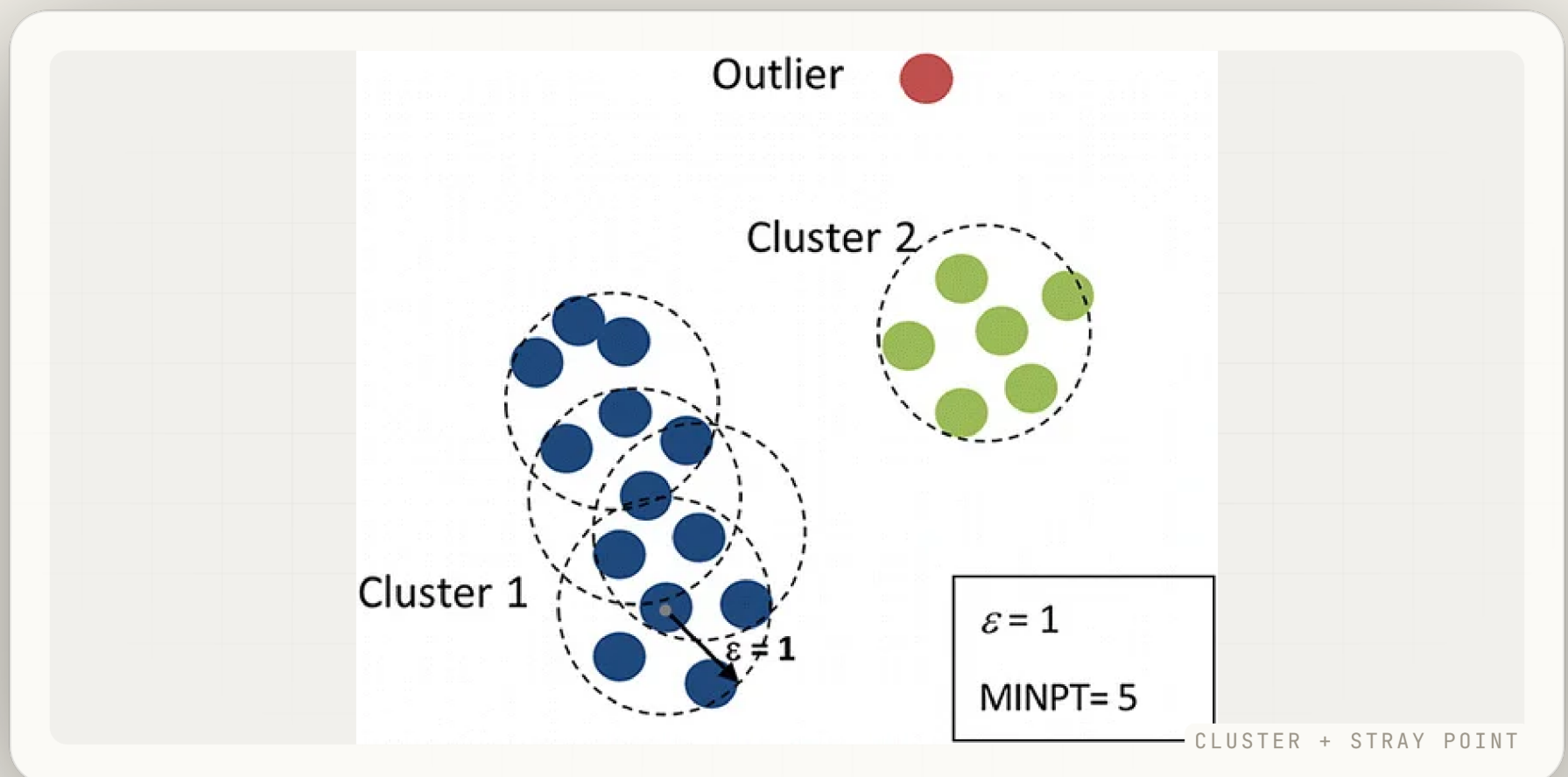
Outliers

the points that don't fit

Data points that differ sharply from the rest, they can **skew model training or signal anomalies**, fraud, or errors worth investigating.

USED IN

- Anomaly detection
- Fraud
- Data cleaning



OPTIMIZER / OPTIMIZATION • TRAINING

Optimizer

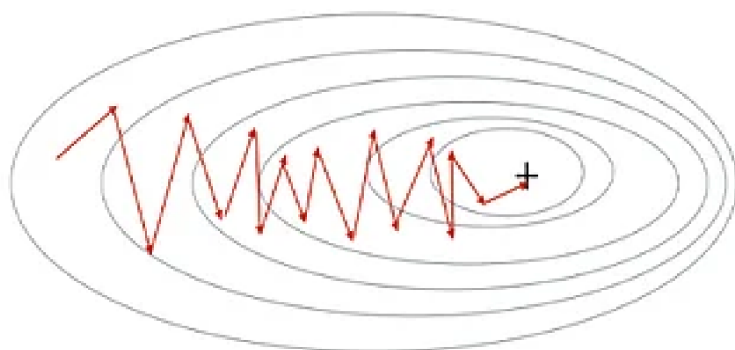
Adam • SGD • descend the loss

The algorithm that **adjusts a model's weights during training** to minimize the loss function, Adam and SGD are the workhorses behind nearly every model.

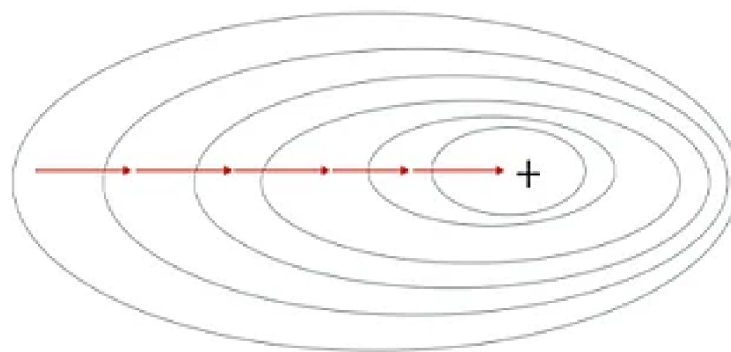
USED IN

- Training loops
- Backpropagation
- Fine-tuning

Stochastic Gradient Descent



Gradient Descent



LOSS SURFACE → DESCENT

ORDINARY DIFFERENTIAL EQUATION • MATH

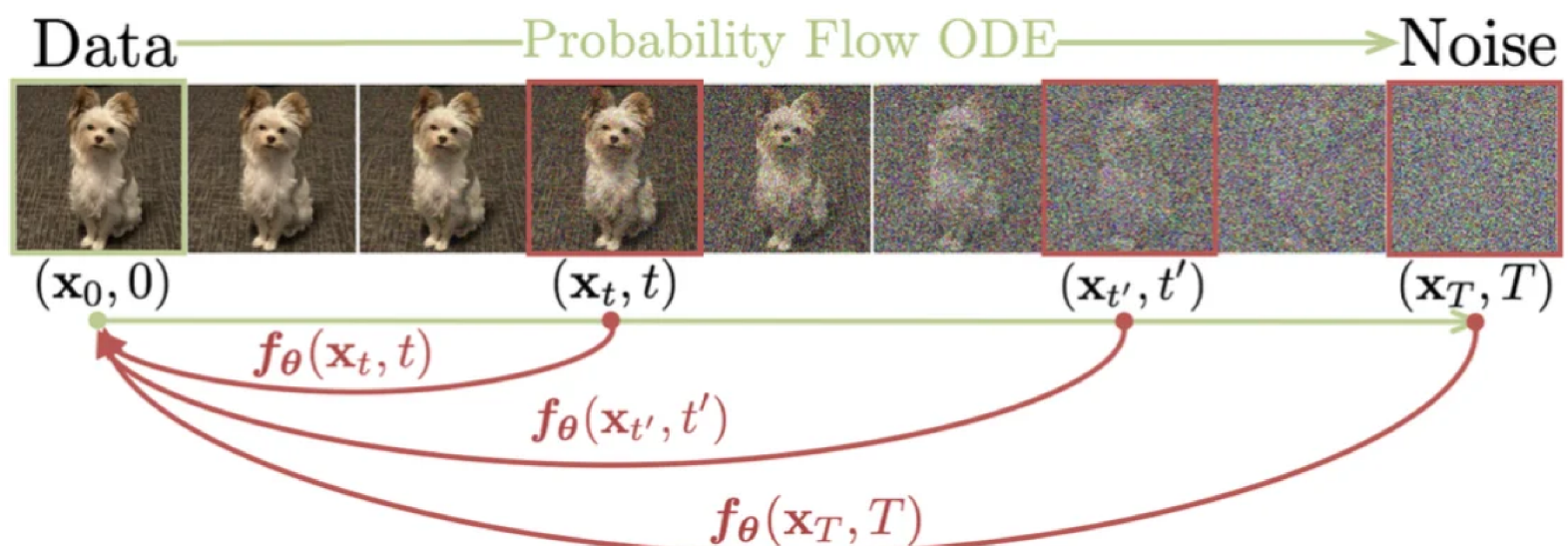
ODE

how a quantity changes over time

An equation describing **how a quantity changes continuously over time**, the math behind Neural ODEs, diffusion models, and continuous-time generation.

USED IN

- Neural ODEs
- Diffusion models
- Physics-informed ML



CONTINUOUS TRAJECTORY

ORTHOGONALITY • EMBEDDINGS

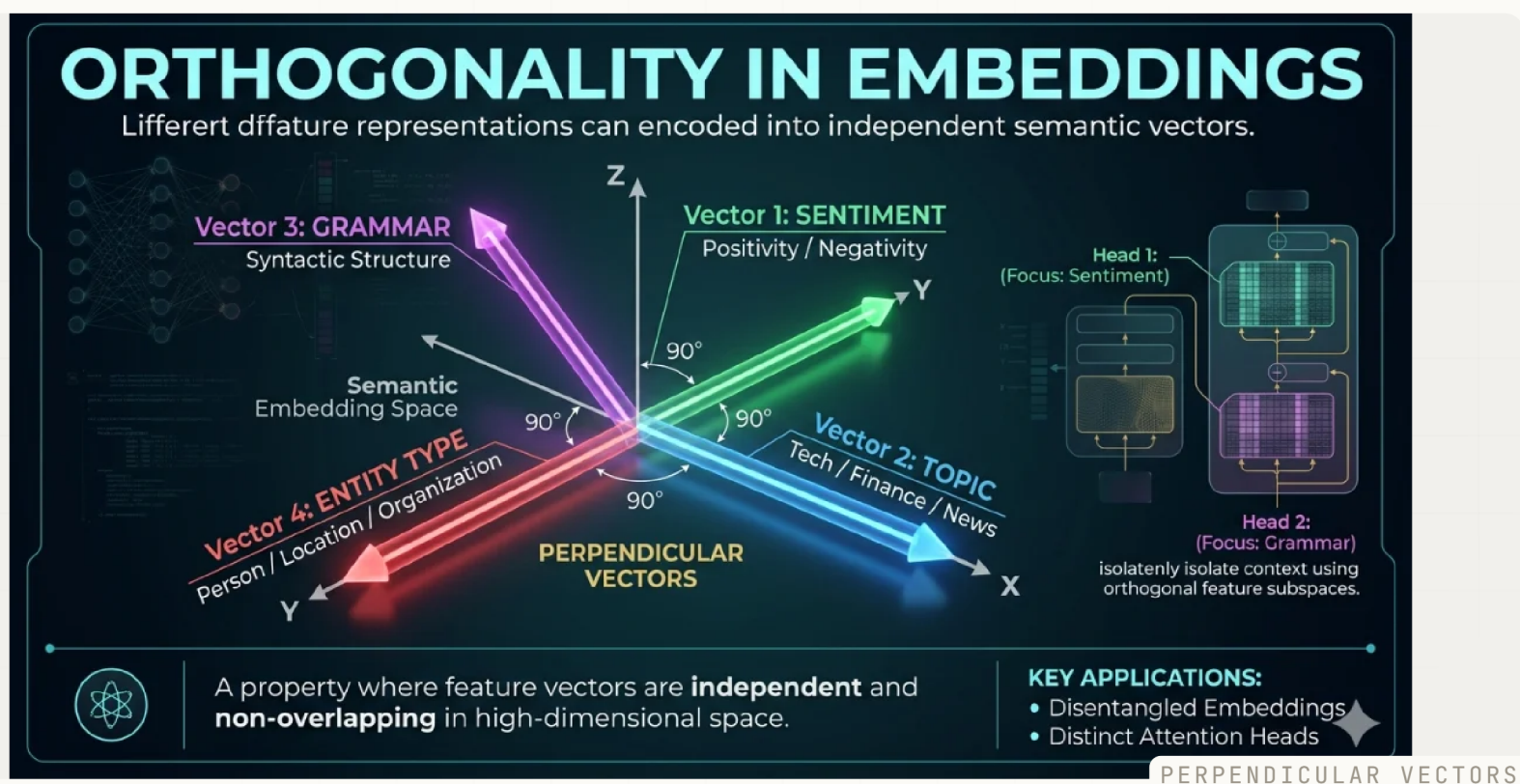
Orthogonality

independent • non-overlapping signals

A property where vectors are **independent and uncorrelated**, relevant to disentangled embeddings and how attention heads capture distinct types of information.

USED IN

- Embeddings
- Attention heads
- Disentanglement



ONE-SHOT PROMPTING • PROMPTING

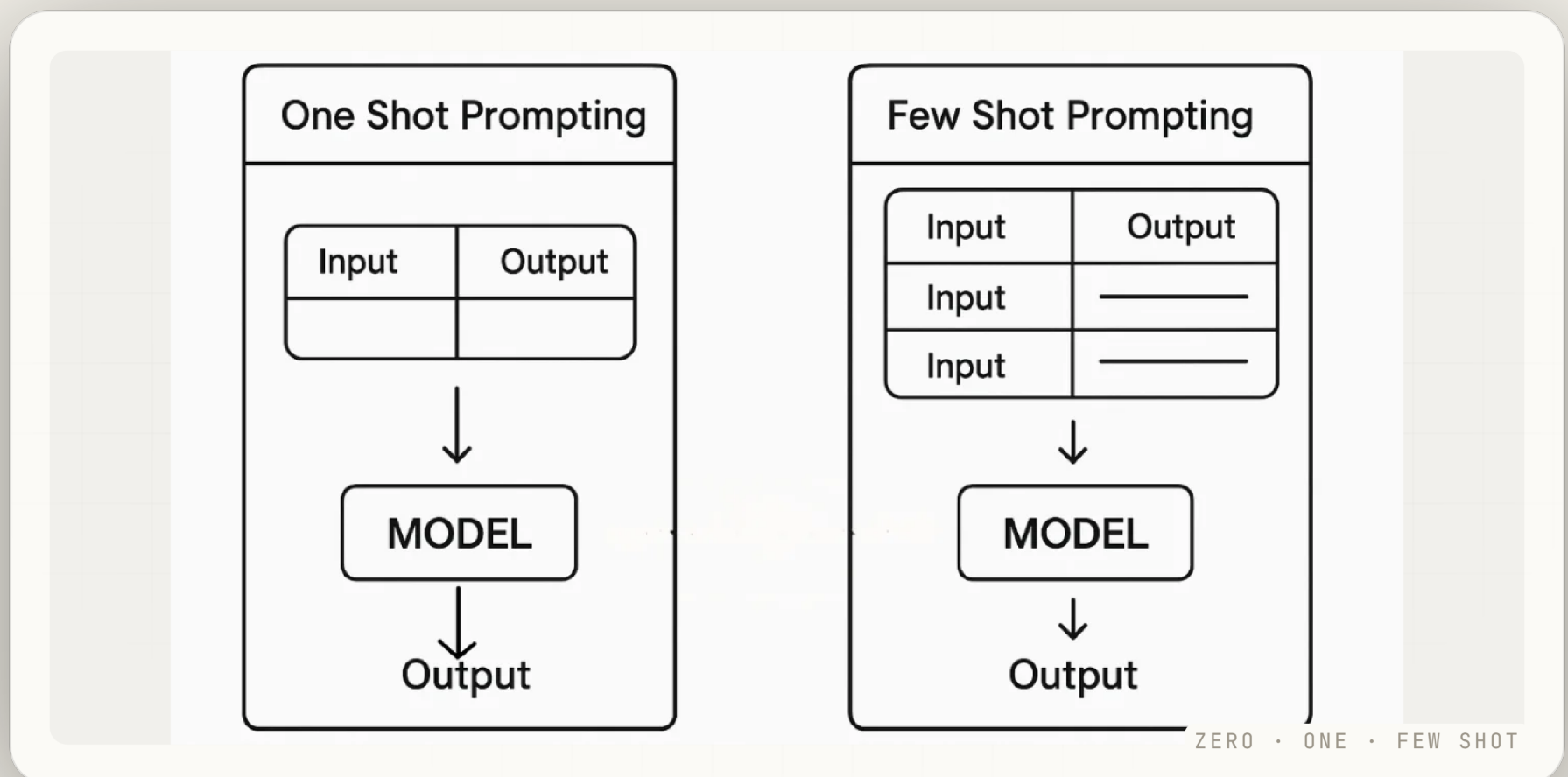
One-shot Prompting

one example • then the task

Giving a model **exactly one example** before asking it to perform a task, just enough to infer the expected format or pattern.

USED IN

- Prompt engineering
- In-context learning
- Few-shot setups



OPEN-WEIGHT MODELS • MODELS

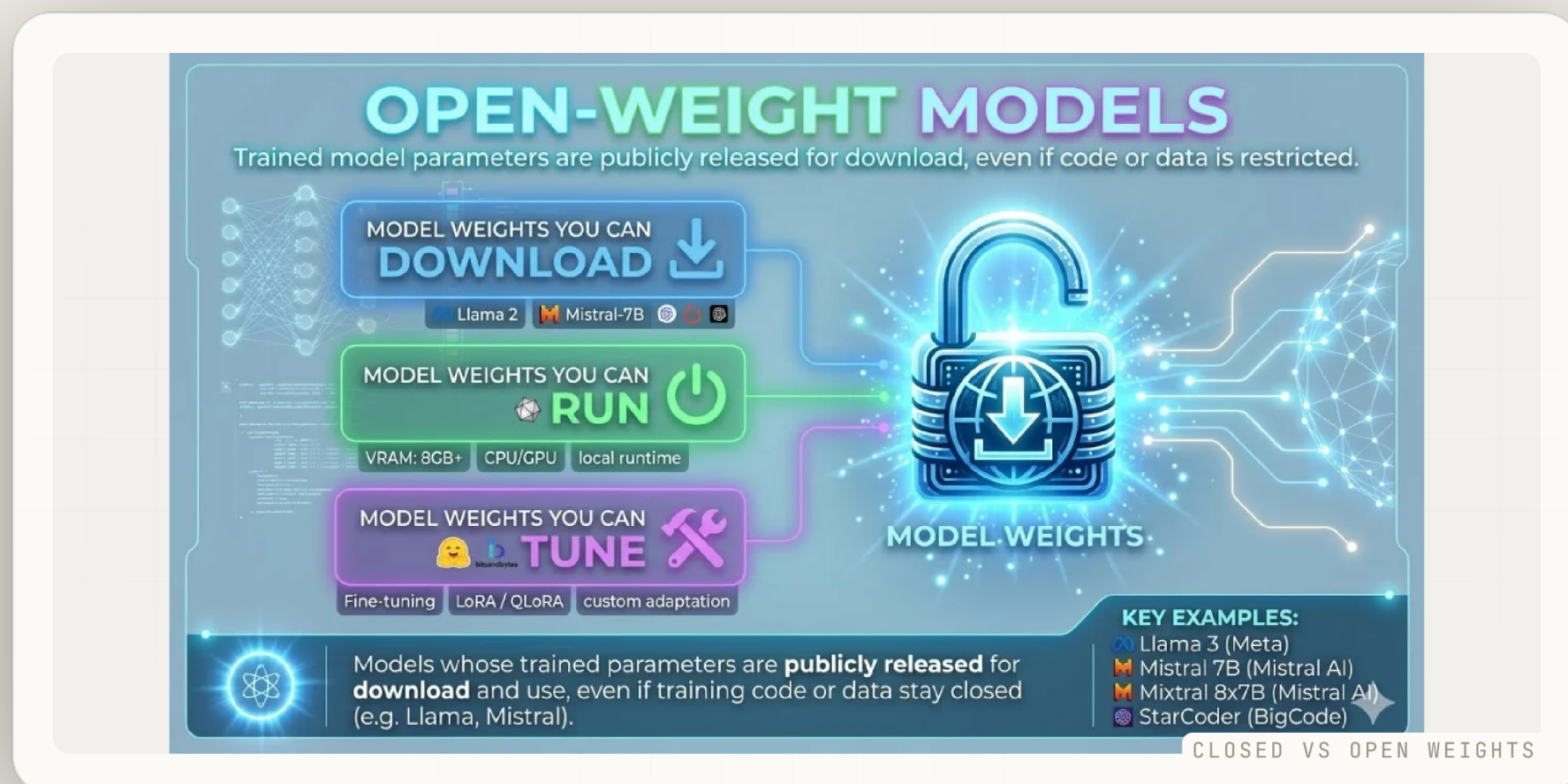
Open-weight Models

weights you can download • run • tune

Models whose **trained parameters are publicly released** for download and use, even if training code or data stay closed (e.g. Llama, Mistral).

USED IN

- Local inference
- Fine-tuning
- On-prem AI



OPTICAL CHARACTER RECOGNITION • VISION

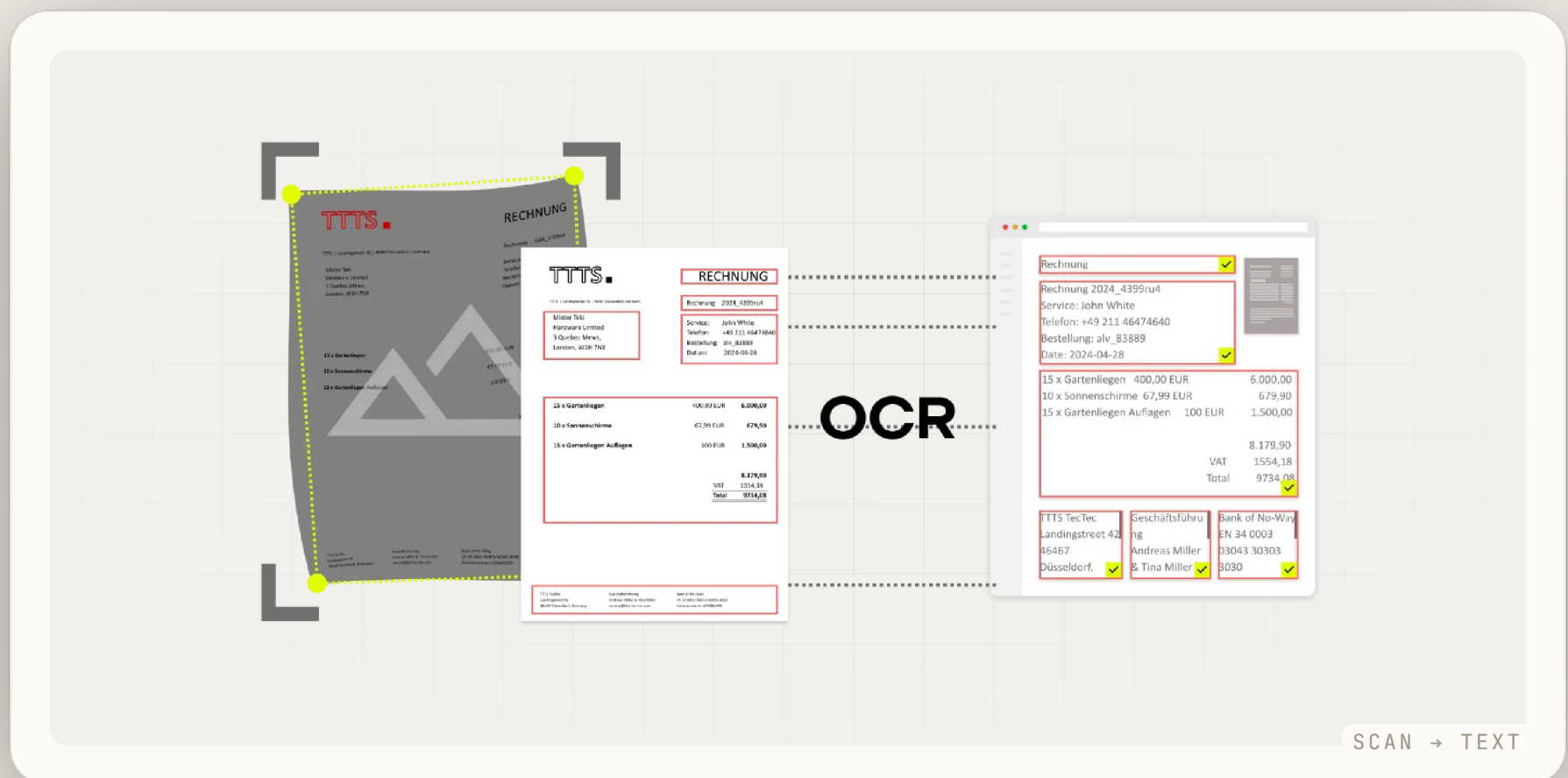
OCR

pixels of text → machine-readable text

Technology that **extracts machine-readable text from images** or scanned documents, often the first step in document AI and RAG pipelines.

USED IN

- Document AI
- RAG ingestion
- Data entry



ORCHESTRATION • AGENTS

Orchestration

what runs when • in what order

Coordinating multiple agents, tools, or pipeline steps into a working system, **deciding what runs when** and how outputs pass between steps.

USED IN

- Agent systems
- LLM pipelines
- Workflows



STEPS → COORDINATED FLOW

OBSERVABILITY • INFRASTRUCTURE

Observability

see inside • trace everything

Monitoring and tracing what an AI system does internally, inputs, outputs, latency, errors, **critical for debugging LLM apps and agents** in production.

USED IN

- LLM ops
- Debugging agents
- Production monitoring

WHAT AN AI

Observability Tool Actually Does



OVALEDGE

- Trace every prompt and response
- Track token usage and cost
- Detect drift before accuracy drops
- Catch hallucinations in production
- Trace failures to the root cause

TRACES • LOGS • METRICS

↓ UP NEXT • TOOLS & COMPANIES

The O stack.

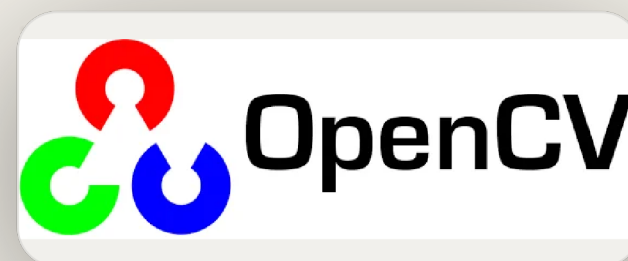
Five O-named tools the AI stack quietly runs on,
from **the library that started computer vision** to
running open models on your own machine.

O

COMPUTER VISION LIBRARY

OpenCV

vision • open-source • everywhere



The world's largest open-source computer vision library, **in roughly 80% of embedded vision projects** since 2000, maintained by the Open Source Vision Foundation.

2000

Vision library since day one

~80%

Of embedded vision projects

Open

Maintained by OSVF non-profit

WHAT IT IS

Open-source computer vision toolkit.

IN AI STACKS

Preprocessing, detection, and robotics.

WHY DEVS USE IT

Battle-tested vision primitives, free.

FIND THEM

opencv.org

LOCAL LLM RUNTIME

Ollama



run open models • on your machine

An open-source tool for downloading and **running open-weight LLMs locally**, Llama, Mistral, Gemma, DeepSeek, with a REST API and CLI. Privacy-first, no cloud.

Local

Runs models on your own hardware

REST

Simple API and CLI

Open

Llama, Mistral, Gemma, DeepSeek

WHAT IT IS

Local runner for open-weight LLMs.

IN AI STACKS

Local dev, prototyping, on-prem apps.

WHY DEVS USE IT

Private, offline model inference.

FIND THEM

ollama.com

LLM API ROUTER

OpenRouter



400+ models • one endpoint

A unified API that **routes requests across 400+ LLMs** from different providers, handling fallback, billing, and model comparison in one endpoint.

400+

Models across providers

One API

Swap models without code changes

Fallback

Auto-routing and billing

WHAT IT IS

Single API across many LLM providers.

IN AI STACKS

Provider-agnostic LLM access layer.

WHY DEVS USE IT

Compare and switch models instantly.

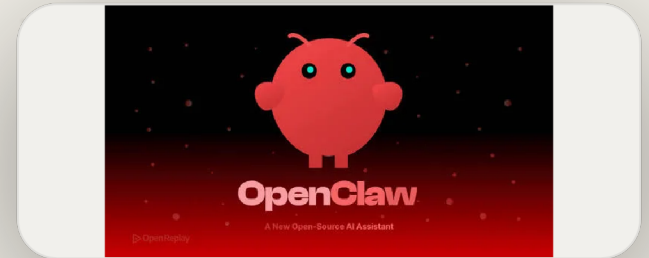
FIND THEM

openrouter.ai

PERSONAL AI ASSISTANT

OpenClaw

self-hosted • 24/7 automation



An open-source, self-hosted personal AI assistant that **connects LLMs to your apps, files, and tools** for round-the-clock automation. Built by Peter Steinberger.

Self-host

Runs on your own infrastructure

24/7

Always-on personal automation

Open

Open-source "Molty" assistant

WHAT IT IS

Self-hosted LLM-powered assistant.

BUILT BY

Peter Steinberger (PSPDFKit founder).

WHY DEVS USE IT

Wire AI into messaging and files.

FIND THEM

github • OpenClaw

LOCAL-FIRST NOTES

Obsidian



plain-text • plugins • your data

A local-first Markdown note app that **stores notes as plain text on your device**, AI features like RAG chat and semantic search arrive via community plugins.

Local

Plain-text files you own

Plugins

RAG chat, semantic search

Markdown

Portable, future-proof notes

WHAT IT IS

Local-first Markdown knowledge base.

AI VIA

Smart Connections, Copilot plugins.

WHY DEVS USE IT

Own your notes; bolt on AI later.

FIND THEM

obsidian.md

↓ UP NEXT • PEOPLE TO FOLLOW

The O mind.

One O-named researcher **worth a follow**, among the most cited names in modern AI, with a hand in nearly every DeepMind landmark.

O

VP OF RESEARCH, DEEPMIND

Oriol Vinyals

Google DeepMind • @OriolVinyalsML



VP of Research and Deep Learning Lead at Google DeepMind and a Gemini co-lead, **one of the most cited AI researchers alive**, behind AlphaStar, seq2seq, and more.

Gemini

Co-lead at Google DeepMind

seq2seq

AlphaStar, AlphaFold, WaveNet

Cited

Among the most cited in AI

KNOWN FOR

AlphaStar, seq2seq, and Gemini.

BEST FOR

Frontier research direction and depth.

WHY FOLLOW

Signals where deep learning heads next.

FIND THEM

[linkedin.com/in/oriol-vinyals-00b3366](https://www.linkedin.com/in/oriol-vinyals-00b3366)

The O stack, in one card.

Save it. Drop a comment with the O-term you use most.

01 **Overfitting**

foundations

02 **Outliers**

data

03 **Optimizer**

training

04 **ODE**

math

05 **Orthogonality**

embeddings

06 **One-shot Prompting**

prompting

07 **Open-weight Models**

models

08 **OCR**

vision

09 **Orchestration**

agents

10 **Observability**

infra

+ **OpenCV • Ollama • OpenRouter**

tools

+ **OpenClaw • Obsidian**

tools

@ **Oriol Vinyals**

people



FOLLOW FOR THE FULL A TO Z

Prasanna Kulkarni

in/prasanna-kulkarni-032650180

P Next